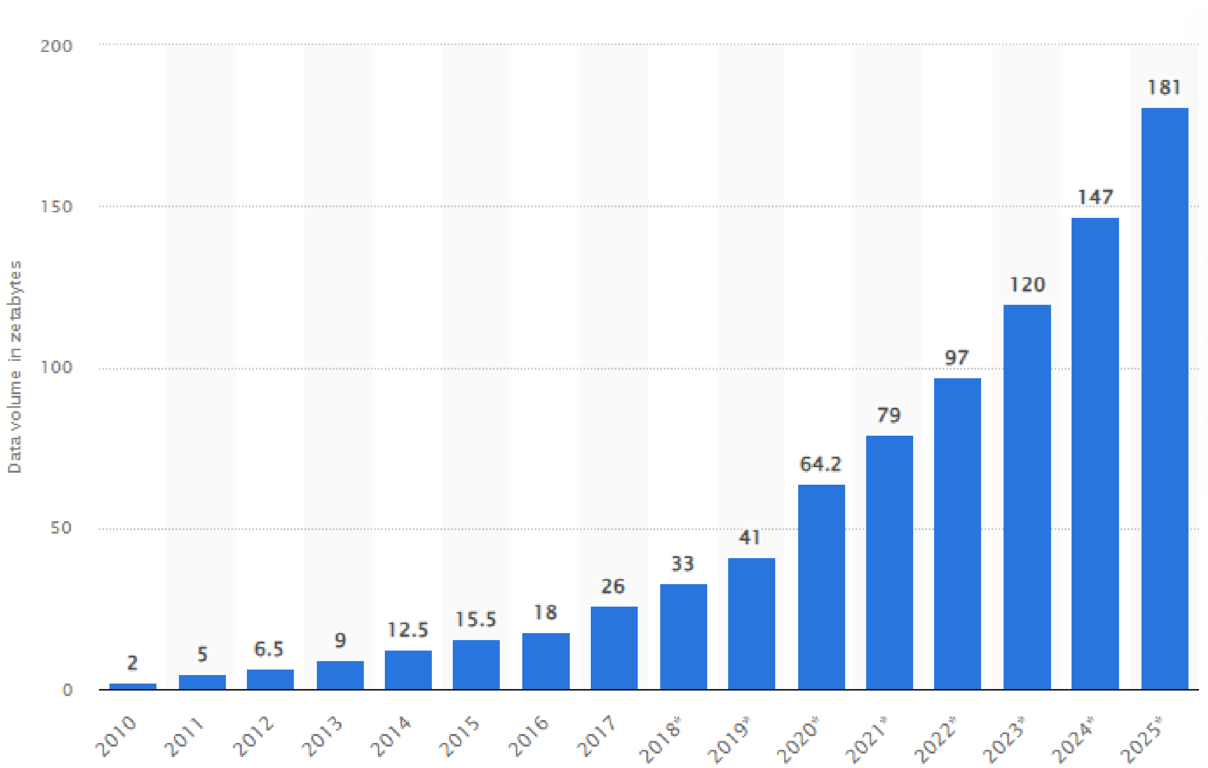
Data Privacy and User Data Control: A New Zealand Perspective

Contents

Chapter 1. Introduction

1.1. Background

In the modern information era, the topic of data privacy and user data control is highly contentious. Data or information is now in ways considered a commodity much like water or natural gas (Elvy, 2018). By 2025 it is forecast that global data creation will grow to more than a total accumulation of 180 zettabytes however, consumers lack control, visibility and awareness in how their data is shared and used (Tsesis, 2014). These factors combined contribute to a mindset in consumers fostering uncertainty and distrust in consumers.



(Taylor, 2023)

Throughout this report we will be reviewing a collection of peer reviewed papers on the topic of Data privacy and user data control. The focus here is on the reoccurring themes of data ownership and sovereignty, privacy and data protection and ethical concerns for data use. The relevance of this topic in today's information-driven world cannot be overstated, as it directly impacts individuals' autonomy and the ethical use of data by corporations and governments.

1.2. Objectives

The primary objective of this literature review is to observe current research that has been conducted on data ownership and sovereignty to truly understand the meaning of these concepts.

We will investigate legal, technological and ethical frameworks that could enable consumers to obtain greater ownership and control over their data in New Zealand whilst comparing this with global trends.

Additionally, we will investigate key factors in data privacy, protection and how organisations and individuals work with data to maintain these principles.

1.3 Scope and Boundaries

1.3.1 Time Frame

This review will focus on the past 15 years of research in this field. Primarily from 2009-2024 to review the most recent trends in the topics outlined above keeping findings relevant and current. The scope is not strictly refined to this as this field has been slowly developing since the start of the information era.

1.3.2 Geographic Constraints

The primary geographic focus of this will be on a global scale however references will be made back to New Zealand where relevant, particularly considering how local laws (e.g., Privacy Act 2020) and cultural factors (e.g., Māori data sovereignty) impact consumer data ownership.

1.3.3 Types of Sources

Peer reviewed Articles: Scholarly articles on data privacy, consumer rights and data monetisation.

Government Reports and Legislation: Documents such as New Zealand's Privacy Act 2020, and reports from government bodies or privacy advocates

Industry reports and white papers: Insights from industry leaders on data ownership models, monetisation and consumer awareness efforts.

Case Studies: Examples of business that have implemented data ownership or monetisation strategies.

Chapter 2. Literature Review

2.1. Overview

Data privacy and user data ownership is a critical topic in today’s world particularly because large multinational companies are now dealing with data across borders. This review will focus on the reoccurring themes of data ownership and sovereignty, privacy and data protection, and ethical concerns in data use that emerge in papers on data privacy and consumer data ownership. These papers have all been peer reviewed and published over the past 15 years and are generally some that are frequently cited by other researchers.

2.2. Data Ownership and Sovereignty

Review and Synthesis

The reviewed papers all emphasise the importance of data ownership and sovereignty and control especially in todays globalised digital economy where data is constantly crossing borders, but researchers agree that legal frameworks and organisations approaches to governing this remain unclear. It is mostly agreed that the concept of data sovereignty refers to the notion that data is subject to the laws and governance structure of the nation where it is collected or stored however as mentioned previously when data transcends borders these structures become less effective especially for large multinational companies who must navigate multinational laws and legislation when transferring, storing or collecting data.

First, when approaching this topic, we must define what data sovereignty is. When the word is broken down sovereignty means “supreme power or authority” but what does this mean in the context of data? Researchers have defined frameworks to classify data sovereignty as well as synthesised other research discussing how varying official regulatory frameworks impact the ability of both organisations and individuals to control their data across borders. One way that this can be facilitated is by classifying the person who is the subject of the data as the sovereign data owner (Hummel et al., 2021).

Even if we classify the sovereign data owner this still poses questions as to who really owns the data. Many researchers have explored the overarching question of data ownership arguing for clearer definitions and legal frameworks particularly from government to support individuals’ rights over their data. Without these clear definitions and frameworks individuals and organisations are at risk of data misuse. The paper “Data Ownership Who Owns My Data” summarises these points and concludes that current frameworks are not viable to facilitate this (Al-Khouri, 2012). Expanding on this point further, local frameworks especially indigenous frameworks such as Māori data sovereignty in New Zealand must be honoured as failure to do so can lead to major consequences for organisations. This raises more questions regarding the implementation these types of frameworks on a global scale (Carroll et al., 2019; Lilley et al., 2024).

Furthermore, researchers pose the question; how we can use emerging technologies to safeguard digital sovereignty arise (Hojati et al., 2021; Zhao et al., 2022). A paper “MapSafe: A complete tool for achieving geospatial data sovereignty” introduces a tool which enables sovereign data owners to obfuscate their data particularly their geospatial data on the web by storing actual data on decentralised file storage and then the corresponding hash on a blockchain showing in practice how technology can help individuals maintain data sovereignty, ensuring that data remains under the control of the rightful or sovereign owners (Sharma et al., 2023).

Finally, it is generally theorised that a global standard for ownership needs to be created to address challenges created with cross border data flows. a concept that has been investigated across many reviews, papers and journals but with global jurisdiction and fundamental differences in national policy, legislation and ethical considerations this has been a contentious issue which poses many challenges in finding common ground (Al-Khouri, 2012; Fischli, 2024; Hummel, Braun, Tretter, et al., 2021; Lilley et al., 2024).

Conclusion

In conclusion from the reviewed papers, we can see that first, we must define data sovereignty as this creates commonality in understanding. Frameworks have been developed particularly in the space of geopolitics which help define data sovereignty (Hummel, Braun, Tretter, et al., 2021). The consensus is that the subject of the data is the sovereign data owner. Once we define sovereignty, we need to look at how we hand data ownership and control over to sovereign data owners whilst still allowing organisations that collect data to manage this. There are still many questions and theories regarding how this is done, and some technologies have been created which help accomplish this but there is still much room for global frameworks and tools to develop in this area (Al-Khouri, 2012).

Strengths and Limitations

Strengths: All papers rely on and effectively emphasise the importance of defining data sovereignty. This establishes common ground for laws to govern data within the jurisdiction that it is collected, stored or transferred in to apply (Hummel et al., 2021). Additionally, researchers highlight the need for clearer legal frameworks to protect data ownership (Al-Khouri, 2012) particularly the governments intervention to safeguard individuals' rights over data. Tools like MapSafe demonstrate how emerging technologies can be used to secure data sovereignty and offer practical applications for protecting geospatial data. Overall, the literature emphasises the need for global standards to manage cross border data flows with consistent advocacy for creating universally applicable frameworks.

Limitations: The definition of data sovereignty tends to be narrow or inconsistent across different papers, as they largely depend on the national legislation from the area where sovereignty is referenced. This lack of a universally agreed upon definition particularly when data crosses borders is a critical shortcoming. Additionally, most papers fail to address how emerging technologies disrupt traditional notions leaving a gap in the understanding of how sovereignty evolves in a digital context.

Agreement, Disagreement, and Inconsistencies

Agreement: Overall, researchers throughout these papers agree that data sovereignty should empower individuals who are the sovereign data owners to control their own data and that currently existing legal frameworks do not suffice. There is also a consensus that a global standard for data ownership would be beneficial, even though it remains difficult to pinpoint due to geopolitical and legal barriers.

Disagreement: One main area of disagreement is what role governments versus organisations hold. Some researchers argue that governments should hold jurisdiction over data and decide how it should be used (Al-Khouri, 2012) whereas others argue that organisations should have freedom to decide. The papers also differ in how they approach the issue of technological solutions. Some are optimistic about this, such as the implementation of blockchain to protect geospatial data where others express scepticism about the readiness of such technologies for widespread adoption.

Inconsistencies: There is much inconsistency when defining data owners. While some papers introduce the notion of a sovereign data owner (Hummel, Braun, Tretter, et al., 2021) other researchers do not always align with this terminology or perspective which leads to varied interpretations of who ultimately holds control over data. Additionally, while all sources recognise the difficulties that surround cross border data flow there is little consensus on how to manage this with some advocating for regional frameworks i.e. (GDPR) while others push for broader coverage on a global scale.

2.3. Privacy and Data Protection

Review and Synthesis

The topic of privacy and data protection is of great importance when discussing data in general particularly when it comes to data governance. The reviewed papers all collectively highlight the challenges in balancing the usage of data to innovate within organisations while still protecting and maintaining privacy over the data. As data continues to cross borders the requirements for frameworks that enforce privacy and safeguard data becomes increasingly prevalent. However, as researchers across the studies reviewed have observed, the effectiveness of current regulatory standards remains a topic of significant debate. In the context of New Zealand, generally data is governed by the privacy act 2020 however it is not just this alone, organisations also need to consider other legislation and frameworks such as Māori data sovereignty (Lilley et al., 2024) and global legal frameworks such as the GDPR. The GDPR in conjunction with other frameworks of this nature have made significant progress in the space of privacy in our digital economy (Jernite et al., 2022; Labadie & Legner, n.d.) but there is a growing consensus that these laws are still insufficient. For instance, the GDPR's "Right to be forgotten" does offer individuals the right to request data deletion but compliance from organisations is slow and inconsistent, particularly when data is being used for advertising and marketing (Tsesis, 2014).

In exploring these various frameworks, there is an emphasis that privacy laws must evolve to keep pace with the acceleration of technology and the changing nature of data collection (Hummel, Braun, & Dabrock, 2021). Much like the definition of digital sovereignty, current privacy protection laws are inconsistent on a global scale creating gaps in enforcement which in turn exposes individuals to potential misuse of their personal data. Additionally, with some methods of anonymising data, researchers still debate whether this is effective enough. Researchers have found that while anonymisation can be effective it often fails to fully protect individuals' identities in the face of advanced data mining techniques where methods using k-anonymity, -diversity, t-closeness, are circumvented by cross referencing data with other big datasets (Zigomitros et al., 2020). Alternatively, many researchers argue that technological tools may provide a sound solution to the issue of data privacy, weather using cloud computing or blockchain, several solutions have been explored (Khan & Hamlen, 2012; Zigomitros et al., 2020)

Furthermore, the concept of data protection is investigated specifically in the context of when large multinational tech companies such as GAFAM (Facebook, Alphabet, Apple, Meta, Microsoft, Etc.) dominate the data economy with their sheer size and granular level of integration with our daily lives. Prainsack argues that privacy, which is often conceptualised as an individual right is inadequately protected in systems where data is commodified and traded by these large companies without sufficient oversight (Prainsack, 2019). Ultimately, this discussion is about balancing the optimisation of profits and improving technological tools whilst still maintaining privacy. The argument here is not that these practices are inherently malicious but that oftentimes there are unwanted side effects of using these large datasets to optimise business (Mehta et al., 2022).

Conclusion

In conclusion, these papers all summarise the need for data privacy and stress its importance. However, researchers all agree that there is an inherent risk to individuals' personal data privacy and that current frameworks and legal structures that govern data protection are oftentimes not sufficient. Again, we see the issue of cross border data flow becoming a prevalent cause of inconsistency, particularly in the case of large GAFAM tech companies given their business is global by nature. Moreover, there is still much progress to be made around data anonymisation and its effectiveness, while these methods are useful, they are not always as reliable as intended. Overall, there is an agreement among researchers that global standards need to be set for data protection and individuals' privacy however this needs to be first addressed on a nation-by-nation basis.

Strengths and Limitations

Strengths: One of the major strengths of the reviewed papers is the diverse methodological approaches. From case studies, legal reviews, to quantitative analysis many different approaches have been made which allow for a highly multi-faceted exploration of this topic. Some papers provide concrete examples of how privacy laws function in practice, highlighting the gaps in enforcement which may not be prevalent in theoretical discussions (Tsesis, 2014). Similarly, more technical evaluations offer truly quantifiable insights into the effectiveness of privacy preserving technologies, contributing much needed verifiable data to the debate. By combining methodologies mentioned above these papers cover a broad spectrum of issues related to privacy, from regulatory shortcomings to technological limitations.

Limitations: Despite its strengths this literature suffers from numerous limitations. Firstly, the case study approach, while useful in highlighting specific examples, often lacks a wide scope of applicability. For example, papers that focus on a limited number of cases may not represent broader issues faced across different sectors. Moreover, while the quantitative analysis in the privacy preserving papers is valuable, it tends to focus on the technical aspects of anonymisation without considering broader legal and ethical Implications. Finally, there is an inherent limitation to the theoretical nature of some discussions. Some papers present ideal theories around data privacy and protection however do not consider the true implications that arise when implementing these in practice.

Agreement, Disagreement, and Inconsistencies

Agreement: Across the reviewed literature there is a consensus that global legislative frameworks such as the GDPR provide essential governance over data (Labadie & Legner, n.d.) but still have shortcomings in their current form. It is widely agreed that while these laws are a step in the right direction their implementation and enforcement remain inconsistent, particularly on a global scale. Global cooperation is necessary to address the challenges posed by cross border data flows. Additionally, many researchers agree the privacy protections need to evolve to address the limitations of anonymisation techniques and other privacy preserving technologies.

Disagreement: In general, research has been inconclusive when deciding the best approach to governing data privacy. Some researchers argue for better legislation and regulatory measures whereas others argue for a more technical approach. For example, one paper focuses on the need for stronger legal frameworks (Hummel, Braun, & Dabrock, 2021) contrarily others focus on privacy preserving technologies to protect individuals (Zigomitros et al., 2020). This difference shows the ongoing debate of legal versus technology-based solutions to this issue.

Inconsistencies: One inconsistency in the reviewed papers is how the issue of privacy is addressed, some researchers address it as an individual and others treat is as collective. For example, some papers state that privacy should be treated as a collective matter (Prainsack, 2019). Other papers that were reviewed opt to treat privacy in a more traditional individual manner where individuals can give consent to their own data and allows for governance of privacy by standard frameworks. This difference shows that there is a wider disagreement between researchers across the reviewed literature regarding how individual’s data is protected in today’s digital age.

2.4. Ethical Concerns in Data Use

Another issue that has emerged from this review is how data is used ethically. Today, public and private organisations alike are using large datasets that contain individual's personal information. The increase in practices of this nature has raised many ethical questions. The reviewed papers highlight the ethical issue of how the data that these organisations collect is used in a responsible manner. Researchers suggest that the datasets need to be used on a way that not only protects the individuals that provide the data but also allows businesses to innovate and move technology forward. Another issue here lies with the industry of AI and Machine learning specifically how these models are trained in a responsible and ethical manner.

A central theme in the literature is in the informed consent process. Consent is fundamental to ethical data use, yet researchers argue that the current consent processes fall short in terms of true informed decision making. With digital tools becoming an integral part of everyday life, every day more and more data is collected on an individual. This poses the question; does an individual really know the extent to which their personal data is being collected? (Zigomitros et al., 2020). Researchers emphasise the ensuring meaningful consent becomes more challenging especially in public health where data collection often happens without direct consent due to regulatory exceptions such as the health insurance portability and accountability act (HIPAA), which permits certain data collection activities without prior authorisation for public health purposes (Annas, 2003; Lee & Gostin, 2009)

Another ethical concern is the issue of data bias and its impact on fairness. AI and ML systems which rely on large datasets to make predictions or decisions, can inherit biases that are present in the data used to train them. This can lead to discriminatory outcomes, especially in areas such as healthcare, criminal justice, and financial services, where algorithm-based decision making can have grave consequences for individuals. One paper discussed an instance where a widely used healthcare algorithm was biased against black patients leading to an inequality in medical care (Obermeyer et al., 2019).

The concept of bias is closely tied to the commercialisation of data in the era where data is now a valuable asset that companies often trade and profit from without individuals' clear consent or knowledge about how their data is being used or shared by corporations like GAFAM (Google/Alphabet Inc., Apple Inc., Facebook Inc., Amazon.com Inc., Microsoft Corp.). Scholars have explored how these major players in the tech industry have structured their business models around collecting and analysing consumer data extensively for purposes such, as targeted advertising and other revenue generating activities (Prainsack, 2019). Though these actions aren't against the law it leads to concerns about using personal information for profit purposes; some believe that companies should be held accountable, for making sure their data handling doesn't hurt people or violate their rights. For instance, Twitter's data usage and suggested frameworks are areas where these discussions take place (Rivers & Lewis, 2014)

Māori data sovereignty in New Zealand plays a crucial role in shaping the country's overall data governance. It represents the right of Māori to control their data collection, access, and usage in alignment with their cultural values. This is essential to avoid repeating historical issues and colonial dynamics. Researchers emphasise the importance of not only respecting indigenous data sovereignty but integrating indigenous governance principles into organisational strategies in countries like New Zealand (Lilley et al., 2024).

Finally, with the rise of technologies like AI, machine learning, and advanced data analytics, the ethical frameworks that guide these technologies need to evolve. Current methods for anonymisation of data face challenges primarily due to the risk of re identification. Furthermore, as technologies continue to develop and progress in capabilities, ethical frameworks need to prioritise flexibility to address the current concerns and future concerns. Some scholars have proposed creating ethical AI frameworks focusing on transparency, accountability, and fairness to ensure that these technologies don’t exacerbate inequalities or generate new ethical problems.

Conclusion

In summary the ethical challenges that surround data use include privacy, data bias, commodification of data, and honouring indigenous data rights. While regulations like GDPR and HIPAA provide some protections, they often fail to keep up with technological advances in big data, AI, and Machine Learning, particularly shown with issues like bias and anonymisation not being effective, for these reasons many new ethical dilemmas are introduced. Moreover, culturally sensitive frameworks like Māori data sovereignty challenge the idea of universally applicable data governance standards which highlights the need for more localised and inclusive approaches.

Strengths and Limitations

Strengths: The literature reviewed contains a wide range of perspectives on ethical data use. Researchers offer their perspectives from legal, technological and cultural angles. It also provides case studies from real life examples where biases have affected critical algorithms.

Limitations: While the reviewed literature goes into depth with theoretical discussion, it often lacks actionable solutions to address ethical concerns in practice. While it identifies many of these challenges there is less clarity on how organisations can balance innovation with ethical responsibility, particularly with emerging technologies.

Agreement, Disagreement, and Inconsistencies

Agreement: Most researchers agree that informed consent and privacy are central to ethical data use. Furthermore, researchers agree that current practices around consent and anonymisation are inadequate, and that new ethical frameworks are required to address the evolving risks posed by AI and big data.

Disagreement: There is a general disagreement around the best solutions to ethical challenges. Some researchers advocate for stronger regulatory frameworks while others advocate for technical solutions. One example of this is in the MapSafe paper discussing a potential technical solution. This is a tool that enhances privacy and by nature considers ethical considerations (Sharma et al., 2023). Alternatively, if we leave this regulation up to the law it falls upon governments to enforce this on organisations and some researchers agree that this is not a practical approach.

Inconsistencies: One major inconsistency throughout these papers is the treatment of privacy and whether this is treated as an individual or collective right. If it is treated as individual the ethical issue lies primarily in consent and respects individuals' autonomy over their own data. Alternatively, if this is viewed as a collective right, frameworks such as Māori data sovereignty become much more manageable as this places importance on communal rights and collective decision making, primarily from Iwi (Lilley et al., 2024). This collective framing creates new ethical complexities as traditional data governance frameworks often rely on individuals.

Chapter 3. Research Gaps and Questions

3.1. Identification of Research Gaps

The current research highlights a major gap in data sovereignty frameworks, particularly in how they address cross-border data flows in respect to Indigenous data sovereignty. In New Zealand, the issue is especially pertinent due to the deep cultural requirement of Māori data sovereignty frameworks, which emphasises the need for culturally aligned data governance. There is currently a lack of integration between local frameworks that protect indigenous rights and global frameworks such as the GDPR. In the context of New Zealand this lack of integration creates a challenge for both Māori communities and the technology industry in New Zealand, due to regulations that are needed to be followed when working with data across borders. This issue poses a gap in existing research that poses a risk to Māori rights and in general, the ethical use of data within New Zealand organisations.

Solutions to this problem are important within New Zealand particularly, both for Māori communities and organisations within the technology industry. For Māori this is important as it not only protects privacy and ownership but also stands as an opportunity to maintain cultural standards and keep tradition alive in the modern age. Additionally, when considering Māori data sovereignty, data can sometimes be considered Taonga or a treasure, this calls for a greater level of care and consideration to the handling of this data. For organisations this will enable more straightforward navigation of data handling not only when working with Māori data but for other organisations around the world handling indigenous data in general.

There have been many attempts to address data sovereignty both on a global scale and locally within countries such as New Zealand. As it stands now New Zealand could be considered a leader in indigenous data governance. The Māori Data Sovereignty Network or Te Rauranga has been actively advocating for the rights of Māori people to govern their own data in attempts to uphold the principles of Kaitiakitanga or guardianship and rangatiratanga or self-determination, this is a concept that could also be adjusted to fit other indigenous nations too. Alternatively, international frameworks such as GDPR preserve individual rights and have been tested to do so but are not sufficient to deal with the collective rights like Māori data sovereignty has been developed to achieve.

Technological solutions have also been explored in the realm of data ownership. As reviewed throughout this paper, one example of this practical implementation is in MapSafe, a tool which harnesses blockchain technology to achieve geospatial data privacy. Although this is a fair attempt at utilising technology it has not been broadly adopted and is narrow in scope. Additionally, these attempts at using technology have been focused in on the individual approach to data ownership as opposed to the collective view, more prominent in indigenous models.

Furthermore, there is currently not a framework that encompasses both indigenous Māori data sovereignty and global regulations such as the GDPR. For this reason, there is a gap when data crosses international borders and is covered by the privacy act 2020 but Māori data sovereignty is not fully considered and accounted for.

The goal of the research based on this paper is to create a comprehensive framework that combines Māori data sovereignty principles, such as Kaitiakitanga and Rangatiratanga, with international data governance frameworks like the GDPR to address existing limitations. In contrast to earlier research, this study will concentrate on constructing a model that considers both individual and collective rights and provides practical solutions for cross-border data transfers that uphold Indigenous governance.

In addition, this research will aim to integrate decentralized technologies like blockchain, but ensure these technologies are implemented with cultural sensitivity. This strategy will not just concentrate on safeguarding data privacy but also assist Māori communities in retaining authority over their data, even if it is handled outside of New Zealand. This innovative framework may be used as a blueprint for other countries with Indigenous communities, bridging the global governance gap and safeguarding indigenous cultural rights.

This method is closely connected to the Ngā Tikanga Paihere concepts of Kaitiaki, Mauri, and Wairua and is shown in the following ways:

- Kaitiaki or guardianship is important to safeguard Māori data as a valuable taonga or treasure for the well-being of future generations. Through creating a structure that acknowledges Māori as the protectors of their information. This method guarantees that cultural authenticity is preserved in the era of technology.
- Mauri, or life force, emphasises the importance of safeguarding the intrinsic value of data, going beyond its simply economic usefulness. In this context, information is more than just a product; it symbolizes the welfare of societies. My study will attempt to guarantee that data governance frameworks are created to protect the mauri of the data, ensuring that it benefits the Māori people and honours the complete value of the information it holds.

- Wairua or spirituality influences the ethical utilisation of data with respect for Māori cultural values. My goal is to integrate Māori data sovereignty principles into worldwide frameworks, ensuring that Māori spiritual and ethical beliefs are honoured when data is utilised, especially on a global scale.

Ultimately, this study aims to create a more inclusive and ethical data governance model by combining Māori cultural principles with global data sovereignty standards, which can have implications both in New Zealand and globally. This method will not just bridge the existing research gap but also make progress in the endeavour that New Zealand continues to be a leader in ethical data management, safeguarding the rights of Indigenous populations while promoting the nation's involvement in the worldwide digital market.

3.2. Formulation of Research Questions

Research Question 1:

How can Māori data sovereignty principles be integrated into global data governance frameworks to ensure cross-border data flows protect Indigenous rights?

Explanation: This research question addresses the gap between local Indigenous frameworks like Māori data sovereignty and global governance standards such as the GDPR which in general, do not sufficiently account for collective rights and cultural values, leaving Māori data vulnerable when data flows cross borders. The question is scoped to focus on integration strategies, ensuring that the research is measurable by identifying specific principles i.e. Kaitiakitanga (Guardianship), Rangatiratanga (Self-determination) and analysing how they can be implemented in global practices.

Contribution: By exploring this question, the research will advance knowledge by proposing a potential model for data governance that incorporates Indigenous rights into a global framework. This would benefit both New Zealand’s approach to data governance and provide a point of reference for other countries with Indigenous populations, promoting a more inclusive system for global data governance.

Research Question 2:

What role can emerging technologies, such as blockchain, play in supporting the protection of Māori data sovereignty on a global scale with cross border data flows?

Explanation: This question addresses the technological gap identified in the current literature, where tools like blockchain have been proposed but are not fully examined from a collective point of view. The scope focuses on emerging technologies and their potential to secure and manage data in ways that respect both local Indigenous rights and global regulatory requirements.

Contribution: This research will contribute to advancing knowledge by evaluating how decentralised technologies can function within Indigenous frameworks like Māori data sovereignty. It will offer practical solutions for New Zealand's technology industry and Indigenous communities, positioning the country as a leader in developing culturally aware technological tools for data protection. Additionally, it will provide insights into the scalability of these tools for broader global application.

3.3. Significance and Contributions

Addressing the identified research gaps surrounding Māori data sovereignty and cross-border data flows holds significant potential to define how Indigenous data rights are managed on a global scale. By integrating Māori data sovereignty principles with global data governance frameworks, this research can provide a culturally centred model that ensures data protection while respecting Indigenous cultural values.

Enhancing Understanding

Cultural and Legal Integration: It will clarify how local Indigenous frameworks, like Māori data sovereignty, can coexist with global governance standards such as the GDPR. This bridges the gap between individual rights which is often the focus of global frameworks and collective Indigenous rights prevalent in the Māori Data Sovereignty framework, creating a more inclusive approach to data sovereignty.

Technological Solutions: The research will offer insights into how technologies such as blockchain and other emerging technologies can protect Māori data on a global scale, providing practical, scalable solutions for ensuring data control and security across borders.

Practical Implications

Legal and Policy Development: Governments, both in New Zealand and globally, can use the findings to design data governance policies that respect Indigenous rights, ensuring better compliance with cross-border data laws.

Business and IT Sector: The IT industry in New Zealand, particularly organisations working with big data, could adopt blockchain-based solutions to manage and protect data sovereignty in a culturally sensitive manner, ensuring ethical data handling for Indigenous communities.

International Standards: This research could influence the development of global data governance frameworks that account for Indigenous legal frameworks, serving as a a reference model for other countries with indigenous populations.

Novelty and Originality

The research is novel because it integrates Māori data sovereignty, which is a unique, culturally specific governance model, with global frameworks like the GDPR, something that has not been fully explored in the literature. Additionally, the use of blockchain technology as a tool for preserving Indigenous data sovereignty on a global scale is an innovative approach that merges technology with cultural preservation, offering a practical solution that aligns with Māori values.

This combination of cultural frameworks, global governance, and emerging technology brings originality to both the academic and practical fields of data sovereignty.

Chapter 4. Conclusion

4.1. Summary

This literature review highlights the growing importance of data privacy, ownership, and sovereignty in a globalised, information driven world. Key themes include the cross-border management of data, the importance of clear legal frameworks, and the ever growing need to respect Indigenous data sovereignty, particularly in New Zealand. While global frameworks like the GDPR offer some protection, they fail to address the collective data rights emphasised in Māori data sovereignty. The review identifies two major gaps: the integration of Māori data governance into global systems and the use of emerging technologies, such as blockchain, to protect Indigenous data.

The review emphasises the importance of developing a hybrid model that combines Indigenous governance with global frameworks, offering both practical and ethical contributions to data sovereignty. This research could serve as a model for other Indigenous groups and influence global data policy development.

4.2. Implications and Future Directions

This review has broad significance in the field of data governance, particularly for New Zealand, both the Māori community and the tech industry alike by advocating for a culturally inclusive model that accounts for indigenous rights while managing cross border data flows. Moving forward with this research we will explore how these indigenous frameworks, specifically Māori data sovereignty can be aligned with global standards and how technologies such as blockchain can support these frameworks.

This literature review has vast implications in the field of data governance, advocating for a culturally inclusive model that accounts for Indigenous rights while managing global cross border data flows. Future research should explore how Indigenous data frameworks can be aligned with global standards and how technologies such as blockchain can support these frameworks.

Potential Research Avenues

Investigation into how we can combine the principles of Māori data sovereignty with global frameworks such as the GDPR.

Exploring how we can use emerging technologies practically such as blockchain to protect indigenous data when moving it across borders.

Methodological Approach

To investigate this, we may use a combination of analysis over existing frameworks to evaluate how we can combine Māori data sovereignty with them, we may also observe existing case studies from around the world where this has been attempted already. Alternatively, if we are investigating the use of blockchain technology we may also investigate case studies or attempt to actively research the practical applications of blockchain. Some theoretical frameworks which may assist in these endeavours are Legal Pluralism and Postcolonial theories, these theories may provide valuable insight into the dynamics of indigenous and global governance frameworks combining.

Challenges

Some potential challenges of this research may be scaling a blockchain based system for global use and the inherent differences in governance models which make the design of this type of system difficult by nature. Additionally, if we are to combine indigenous cultural values into global frameworks, we are likely to face resistance. This makes cross culture collaboration essential for success. Finally, when investigating and applying Māori data sovereignty in our research we need to ensure that we honour Taonga particularly when handling information and data, and ensure that all appropriate cultural customs are followed, this could involve consulting with local Iwi or discussing with others who are experts on this topic.

References

Al-Khouri, A. M. (2012). Data Ownership: Who Owns "My Data"? *INTERNATIONAL JOURNAL OF MANAGEMENT & INFORMATION TECHNOLOGY*, *2*(1), 1–8. https://doi.org/10.24297/ijmit.v2i1.1406

Annas, G. J. (2003). HIPAA Regulations—A New Era of Medical-Record Privacy? *New England Journal of Medicine*, *348*(15), 1486–1490. https://doi.org/10.1056/NEJMlim035027

Carroll, S. R., Rodriguez-Lonebear, D., & Martinez, A. (2019). Indigenous Data Governance: Strategies from United States Native Nations. *Data Science Journal*, *18*(1), 31. https://doi.org/10.5334/dsj-2019-031

Elvy, S.-A. (2018). *Commodifying Consumer Data in the Era of the Internet of Things*.

Fischli, R. (2024). Data-owning democracy: Citizen empowerment through data ownership. *European Journal of Political Theory*, *23*(2), 204–223. https://doi.org/10.1177/14748851221110316

Hojati, M., Farmer, C., Feick, R., & Robertson, C. (2021). Decentralized geoprivacy: Leveraging social trust on the distributed web. *International Journal of Geographical Information Science*, *35*(12), 2540–2566. https://doi.org/10.1080/13658816.2021.1931236

Hummel, P., Braun, M., & Dabrock, P. (2021). Own Data? Ethical Reflections on Data Ownership. *Philosophy & Technology*, *34*(3), 545–572. https://doi.org/10.1007/s13347-020-00404-9

Hummel, P., Braun, M., Tretter, M., & Dabrock, P. (2021). Data sovereignty: A review. *Big Data & Society*, *8*(1), 205395172098201. https://doi.org/10.1177/2053951720982012

Jernite, Y., Nguyen, H., Biderman, S., Rogers, A., Masoud, M., Danchev, V., Tan, S., Luccioni, A. S., Subramani, N., Dupont, G., Dodge, J., Lo, K., Talat, Z., Johnson, I., Radev, D., Nikpoor, S., Frohberg, J., Gokaslan, A., Henderson, P., … Mitchell, M. (2022). Data Governance in the Age of Large-Scale Data-Driven Language Technology. *2022 ACM Conference on Fairness, Accountability, and Transparency*, 2206–2222. https://doi.org/10.1145/3531146.3534637

Khan, S. M., & Hamlen, K. W. (2012). AnonymousCloud: A Data Ownership Privacy Provider Framework in Cloud Computing. *2012 IEEE 11th International Conference on Trust, Security and Privacy in Computing and Communications*, 170–176. https://doi.org/10.1109/TrustCom.2012.94

Labadie, C., & Legner, C. (n.d.). *Understanding Data Protection Regulations from a Data Management Perspective: A Capability-Based Approach to EU-GDPR*.

Lee, L. M., & Gostin, L. O. (2009). Ethical Collection, Storage, and Use of Public Health Data: A Proposal for a National Privacy Protection. *JAMA*, *302*(1), 82. https://doi.org/10.1001/jama.2009.958

Lilley, S., Oliver, G., Cranefield, J., & Lewellen, M. (2024). Māori data sovereignty: Contributions to data cultures in the government sector in New Zealand. *Information, Communication & Society*, 1–16. https://doi.org/10.1080/1369118X.2024.2302987

Mehta, S., Dawande, M., Janakiraman, G., & Mookerjee, V. (2022). An Approximation Scheme for Data Monetization. *Production and Operations Management*, *31*(6), 2412–2428. https://doi.org/10.1111/poms.13676

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, *366*(6464), 447–453. https://doi.org/10.1126/science.aax2342

Prainsack, B. (2019). Logged out: Ownership, exclusion and public value in the digital data and information commons. *Big Data & Society*, *6*(1), 205395171982977. https://doi.org/10.1177/2053951719829773

Rivers, C. M., & Lewis, B. L. (2014). Ethical research standards in a world of big data. *F1000Research*, *3*, 38. https://doi.org/10.12688/f1000research.3-38.v2

Sharma, P., Martin, M., & Swanlund, D. (2023). MAPSAFE: A complete tool for achieving geospatial data sovereignty. *Transactions in GIS*, *27*(6), 1680–1698. https://doi.org/10.1111/tgis.13094

Taylor, P. (2023, November 16). Data Created Worldwide. *Data Created Worldwide*. https://www.statista.com/statistics/871513/worldwide-data-created/

Tsesis, A. (2014). The Right to Erasure: Privacy, Data Brokers, and the Indefinite Retention of Data. *WAKE FORESTLAW REVIEW*.

Zhao, P., Cedeno Jimenez, J. R., Brovelli, M. A., & Mansourian, A. (2022). Towards geospatial blockchain: A review of research on blockchain technology applied to geospatial data. *AGILE: GIScience Series*, *3*, 1–6. https://doi.org/10.5194/agile-giss-3-71-2022

Zigomitros, A., Casino, F., Solanas, A., & Patsakis, C. (2020). A Survey on Privacy Properties for Data Publishing of Relational Data. *IEEE Access*, *8*, 51071–51099. https://doi.org/10.1109/ACCESS.2020.2980235